

LongDocFACTScore: Evaluating the Factuality of Long Document Abstractive Summarisation

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Abstract

Maintaining factual consistency is a critical issue in abstractive text summarisation, however, it cannot be assessed by traditional automatic metrics used for evaluating text summarisation, such as ROUGE scoring. Recent efforts have been devoted to developing improved metrics for measuring factual consistency using pre-trained language models, but these metrics have restrictive token limits, and are therefore not suitable for evaluating long document text summarisation. Moreover, there is limited research and resources available for evaluating whether existing automatic evaluation metrics are fit for purpose when applied in long document settings. In this work, we evaluate the efficacy of automatic metrics for assessing the factual consistency of long document text summarisation. We create a human-annotated data set for evaluating automatic factuality metrics, LongSciVerify, which contains fine-grained factual consistency annotations for long document summaries from the scientific domain. We also propose a new evaluation framework, LongDocFACTScore, which is suitable for evaluating long document summarisation. This framework allows metrics to be efficiently extended to any length document and outperforms existing state-of-the-art metrics in its ability to correlate with human measures of factuality when used to evaluate long document summarisation data sets. We make our code and LongSciVerify data set publicly available: <https://github.com/jbshp/LongDocFACTScore>.

Keywords: Evaluation Methodologies, Summarisation, Natural Language Generation

1. Introduction

Factual inconsistency, i.e., when a generated summary is not entailed by its source document, is a well-documented limitation of modern neural summarisation methods (Maynez et al., 2020; Wallace et al., 2021). Although Large Language Models (LLMs) have shown greatly superior performance on a range of NLP tasks, including summarisation (Zhang et al., 2023; Xie et al., 2023), and are increasingly being used for summarisation of long documents in real world applications, even the best performing models, such as GPT-4 (OpenAI, 2023), are flawed in their ability to remain factual consistent (Bang et al., 2023; Ye et al., 2023; Min et al., 2023).

Human evaluation is generally regarded as the gold standard for evaluating generative models, yet it is timely and costly to conduct, particularly for tasks involving long documents, and thus only a small proportion of long document summarisation studies perform a human evaluation on long document data sets (Krishna et al., 2023). Consequently, there is a requirement for effective automatic evaluation metrics which align to human judgement in long document settings.

Although ROUGE scoring (Lin, 2004) is the traditional metric for automatic evaluation of text summarisation, it is flawed and does not correlate well with human judgement (Yuan et al., 2021; Huang

et al., 2020; Kryscinski et al., 2019). There have been efforts to develop improved model-based metrics for measuring factual consistency (Scialom et al., 2021; Yuan et al., 2021; Kryscinski et al., 2020; Qin et al., 2022; Fu et al., 2023; Liu et al., 2023), however, the studies proposing these metrics only conduct evaluation on short document summarisation data sets (Hermann et al., 2015; Grusky et al., 2018; Narayan et al., 2018; Pagnoni et al., 2021) and there is limited research or available data sets for evaluating these automatic metrics in long document settings.

Since modern evaluation metrics use pre-trained language models (PLMs), they are only able to process a limited number of tokens at a time and must truncate, on average, over half of the tokens of a long source document in their calculations. Therefore, they cannot be applied effectively when used in long document settings (Koh et al., 2022). This issue is exacerbated when evaluating factual consistency, where many of the metrics are designed to be reference-free (Yuan et al., 2021; Fu et al., 2023; Liu et al., 2023; Scialom et al., 2021; Kryscinski et al., 2020), i.e., they use the source document (generally a much longer document), rather than a gold summary, in their calculations.

In this work, we propose a reference-free evaluation framework, LongDocFACTScore, intended for assessing the factual consistency of abstractive summarisation of long documents. We show

that this framework outperforms all other automatic metrics evaluated in their correlation with human annotations of factuality on the long document data sets in our experiments.

Our proposed framework can be efficiently be extended to any length document and incorporates fine-grained, sentence-level assessments of factuality consistency to give a document-level score for the factual consistency of a summary. We conduct an evaluation of the efficacy and efficiency of LongDocFACTScore and other automatic evaluation metrics on a range of long and short document data sets.

Addressing the scarcity of resources for evaluating automatic metrics in long document settings, we create a long document data set of the scientific domain with fine-grained, expert, human annotations of factual consistency, which we make available alongside our code. We hope that this resource encourages future work into the evaluation of automatic metrics in long document settings.

2. Related Work

2.1. Automatic Evaluation Metrics for Evaluating Factual Consistency

ROUGE scoring (Lin, 2004), which uses word overlap between two texts to calculate their similarity, has long been the popular automatic metric used for evaluation of text summarisation. However, more recently, model-based metrics, such as BERTScore (Zhang et al., 2020b), which measures agreement at a token level between the cosine similarity of BERT-based (Devlin et al., 2019) embeddings, have shown improved correlation with human judgement. Additionally, reference-free model-based metrics have shown improved performance for the evaluation of factual consistency on short document summarisation data sets. FactCC (Kryscinski et al., 2020) uses a fine-tuned BERT-based classifier to predict, for each sentence of a summary, whether it is correct or incorrect, given its source document. QuestEval (Scialom et al., 2021) uses T5-based models (Raffel et al., 2020) for a question generation and answering approach. BARTScore (Yuan et al., 2021) uses BART (Lewis et al., 2020) to calculate the log probability of generating a sequence of text, given a second sequence, to predict an automatic score. T5SCORE (Qin et al., 2022) uses T5-based models and combines the generative approach taken by BARTScore with a discriminative approach - i.e., fine-tuning a model to predict a quality score. Unfortunately, many of these model-based metrics are costly to run, for example, QAGS (Wang et al., 2020), another question-answering based metric, running on a single NVIDIA v100 GPU, will take 4 days to process

the CNN/DM test data set (Nan et al., 2021). Other recent works have proposed the use of LLMs for evaluation of NLP tasks (Fu et al., 2023; Luo et al., 2023; Liu et al., 2023), but still limit their evaluation to short document summarisation data sets.

2.2. Frameworks for Evaluation of Long Document Summarisation

There has been limited research into automatic evaluation metrics for long document summarisation, despite the value of summarisation being derived mostly when applied to long documents. Koh et al., (2022) carried out a survey of long document summarisation and found a gap in research for automatic metrics which could efficiently and effectively be applied to long document data sets. Krishna et al., (2023) propose guidelines for evaluation of long document summarisation, and provide LongEval, the only publicly available long document summarisation data set with human annotations of factual consistency which we could find at the time of conducting our research. SMART (Amplayo et al., 2022) proposes a method of extending evaluation metrics to long documents by cycling through them, but do not propose an efficient way of doing so, nor do they evaluate their work on long document summarisation data sets due to the lack of availability of these resources.

3. Methods

In this section, we describe the LongDocFACTScore framework. This framework builds on existing evaluation metrics but applies them in a novel way, providing a summary-level factuality score that considers fine-grained statements, whilst scaling efficiently though a document of any length. LongDocFACTScore evaluates each sentence in a predicted summary against the most similar sections of a source document, calculated using the cosine similarity of their sentence embeddings (Reimers and Gurevych, 2019). Individual summary sentences are evaluated to ensure fine-grained statements are considered in the predicted summary-level score, whilst sentence embeddings are used to improve efficiency of the framework.

To calculate LongDocFACTScore, both the source document $D = \langle s_i, i \in I \rangle$ and its generated summary $S = \langle s_j, j \in J \rangle$ are split into sentences using the nltk library¹. Splitting a document into sentences before applying an evaluation metric has shown to be effective in prior works (Min et al., 2023; Amplayo et al., 2022). For each of these sentences, sentence embeddings are generated

¹<https://www.nltk.org>

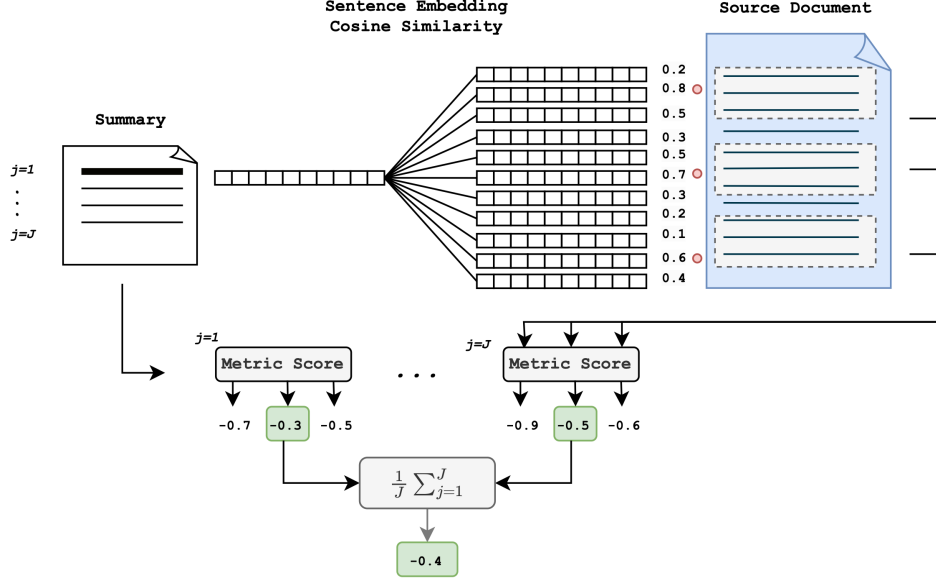


Figure 1: Illustration of the LongDocFACTScore framework.

using the sentence-transformers library² initialised with the bert-base-nmli-mean-tokens model³. For each sentence in the predicted summary s_j , the cosine similarity between its sentence embedding and the sentence embedding of each sentence in the source document s_i is calculated. D is then re-indexed by the cosine similarity scores, so that the new index k is sorted by:

$$\arg \max_{i \in I} (\text{cosine_similarity}(s_j, s_i)). \quad (1)$$

The K most similar source document sentences are then selected and are each concatenated with their preceding and following sentences, thus giving $s_k^* = s_{k-1} + s_k + s_{k+1}$, to create the sequence of slightly longer text snippets. We select the K most similar source document text snippets to improve the efficiency of the framework. We assume the most similar source document text snippets are the ones most likely to be relevant to make an assessment of factual consistency.

The metric score is then calculated between each of the source document text snippets s_k^* and the summary sentence s_j , and the maximum of these scores is taken. In this work, we set $K = 3$, a decision which we justify in Section 5.5.

For each sentence, s_j in S , of the generated summary, the process is repeated, resulting in one score per generated summary sentence. The mean

of these scores is then calculated, providing an overall summary score given by the equation:

$$\frac{1}{J} \sum_{j=1}^J \max_{k=\{1,2,3\}} (\text{metric}(s_j | s_k^*)). \quad (2)$$

Figure 1 illustrates this framework, showing for a single sentence in the generated summary, the similarity scores being calculated for every sentence in the source document, and the resulting three highest scoring sentences being concatenated with their surrounding sentences. A metric score is then calculated between these three source document text snippets and the summary sentence. As indicated in Figure 1, this process is repeated for every sentence in the generated summary and the scores are averaged. For contrast, Figure 2 shows the method for directly applying an automatic scoring metric to a long document, without applying the LongDocFACTScore framework. The entire generated summary and the truncated long source document are directly input to the metric, resulting in one score. Consequently, there are three fundamental differences between LongDocFACTScore and an automatic metric designed for short document evaluation:

- The first difference is that LongDocFACTScore considers sections of text from the full length of the source document in its calculation (using sentence embeddings to select the most relevant from across the document) whereas other metrics truncate the source document. For metrics applied without the LongDocFACTScore framework, if a generated summary includes content from the latter part of

²<https://github.com/UKPLab/sentence-transformers>

³<https://huggingface.co/sentence-transformers/bert-base-nli-mean-tokens>